20 Pat throws a fair coin n times.

Find an expression, in terms of n, for the probability that Pat gets at least 1 head and at least 1 tail.

Exclude all heads and all tails.

$P = 1 - (1/2)^n - (1/2)^n$

$= 1 - 2(1/2)^n$

1 mark
all-heads or all-tails

1 mark
subtract both

$1 - 2(1/2)^n$

(Total for Question 20 is 2 marks)

DO NOT WRITE IN THIS AREA